Exp 18-Splitting a DataFrame into Groups Based on School Code and Class

Aim

To write a Pandas program to split the given DataFrame into groups based on school code and class.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing student details.
3. Use the `groupby()` function with the `school` and `class` columns.
4. Iterate through each group and display the grouped data.
5. Print the groups based on school code and class.

Program

```
import pandas as pd

# Create the DataFrame

data = {
    'school': ['s001', 's002', 's003', 's001', 's002', 's004'],
    'class': ['V', 'V', 'VI', 'VI', 'V', 'VI'],
    'name': ['Alberto Franco', 'Gino Mcneill', 'Ryan Parkes',
             'Eesha Hinton', 'Gino Mcneill', 'David Parkes'],
    'date_Of_Birth': ['15/05/2002', '17/05/2002', '16/02/1999',
                      '25/09/1998', '11/05/2002', '15/09/1997'],
    'age': [12, 12, 13, 13, 14, 12],
    'height': [173, 192, 186, 167, 151, 159],
    'weight': [35, 32, 33, 30, 31, 32],
    'address': ['street1', 'street2', 'street3',
                'street1', 'street2', 'street4']
}

df = pd.DataFrame(data)

# Group by school and class

grouped = df.groupby(['school', 'class'])

print("Groups based on school code and class:\n")

for key, value in grouped:
    print("Group:", key)
    print(value)
    print()
```

OUTPUT

```
Groups based on school code and class:

Group: ('s001', 'V')
  school class           name date_Of_Birth  age  height  weight  address
0   s001     V  Alberto Franco    15/05/2002   12     173      35  street1

Group: ('s001', 'VI')
  school class         name date_Of_Birth  age  height  weight  address
3   s001    VI  Eesha Hinton    25/09/1998   13     167      30  street1

Group: ('s002', 'V')
  school class         name date_Of_Birth  age  height  weight  address
1   s002     V  Gino Mcneill    17/05/2002   12     192      32  street2
4   s002     V  Gino Mcneill    11/05/2002   14     151      31  street2

Group: ('s003', 'VI')
  school class         name date_Of_Birth  age  height  weight  address
2   s003    VI  Ryan Parkes    16/02/1999   13     186      33  street3

Group: ('s004', 'VI')
  school class          name date_Of_Birth  age  height  weight  address
5   s004    VI  David Parkes    15/09/1997   12     159      32  street4
```

Exp 19-Displaying the Shape of the World Alcohol Consumption Dataset and Extracting Column Names

Aim

To write a Pandas program to display the dimensions (shape) of the World alcohol consumption dataset and extract the column names from the dataset.

Procedure

1. Import the Pandas library.
2. Create the DataFrame containing the alcohol consumption data.
3. Use the `shape` attribute to find the number of rows and columns.
4. Use the `columns` attribute to extract the column names.
5. Display the shape and column names.

Program

```
import pandas as pd

# Create the DataFrame

data = {
    'Year': [1986, 1986, 1985, 1986, 1987],
    'WHO region': ['Western Pacific', 'Americas', 'Africa',
                   'Americas', 'Americas'],
    'Country': ['Viet Nam', 'Uruguay', "Cte d'Ivoire",
                'Colombia', 'Saint Kitts and Nevis'],
    'Beverage Types': ['Wine', 'Other', 'Wine',
                       'Beer', 'Beer'],
    'Display Value': [0.00, 0.50, 1.62, 4.27, 1.98]
}

df = pd.DataFrame(data)

# Display shape of the DataFrame

print("Shape of the DataFrame:")
print(df.shape)

# Display column names

print("\nColumn names:")
print(df.columns.tolist())
```

OUTPUT

```
Shape of the DataFrame:
(5, 5)

Column names:
['Year', 'WHO region', 'Country', 'Beverage Types', 'Display Value']
```

Exp 20- Finding the Index of a Given Substring in a DataFrame Column

Aim

To write a Pandas program to find the index of a given substring in a DataFrame column.

Procedure

1. Import the Pandas library.
2. Create a DataFrame with sample data.
3. Select the column in which the substring needs to be searched.
4. Use the `str.contains()` function to find the substring.
5. Extract the index values of the matching rows.
6. Display the index values.

Program

```
import pandas as pd

# Create DataFrame

df = pd.DataFrame({
    'Country': [
        'Viet Nam',
        'Uruguay',
        "Cote d'Ivoire",
        'Colombia',
        'Saint Kitts and Nevis'
    ]
})

# Find the index of rows containing the substring 'Saint'

result = df[df['Country'].str.contains('Saint')].index

print("Index of rows containing the substring:")
print(result.tolist())
```

OUTPUT

```
Index of rows containing the substring:
[4]
```